FIELDS, COMPLETE LOCAL RINGS, THE INTEGERS, CHARACTERISTIC ZERO DEDEKIND DOMAINS AND THEIR FINITE TYPE EXTENSIONS ARE EXCELLENT

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Allowed standard prerequisites (editorial).

- Ubiquity of G-rings and of J-2 rings for these classes. Proposition \ref{proposition-ubiquity-G-ring} and Proposition \ref{proposition-ubiquity-J-2}, cited at proof lines 13398-13399. Supply two of the three ingredients of excellence.
- Cohen-Macaulay rings are universally catenary, and complete local rings are universally catenary. Algebra, Lemma \ref{algebra-lemma-CM-ring-catenary} and Algebra, Remark \ref{algebra-remark-Noetherian-complete-local-ring-universally-catenary}, cited at proof lines 13400-13406. Supply universal catenarity for the classes listed.

Source reference. Original lemma 07QW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9846–9846. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07PX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9812–9812. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9799–9799. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00NM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 864–864. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 032C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4015–4015. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Proposition 0.2. *The following types of rings are excellent:*

(1) *fields,*
(2) *Noetherian complete local rings,*
(3) $\mathbf{Z}$*,*
(4) *Dedekind domains with fraction field of characteristic zero,*
(5) *finite type ring extensions of any of the above.*

Proof. See Propositions 07PX and 07PJ to see that these rings are G-rings and have J-2. Any Cohen-Macaulay ring is universally catenary, see Algebra, Lemma 00NM. In particular fields, Dedekind rings, and more generally regular rings are universally catenary. Via the Cohen structure theorem we see that complete local rings are universally catenary, see Algebra, Remark 032C. □